

### Unique features of Apache Cassandra

| Features                                                  | Description                                                                                                                                   |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Decentralised                                             | Data is distributed across the cluster.<br>There is no master node; every node can service any request.<br>Each node contains different data. |
| Supports replication and multiple data centre replication | It has distributed architecture.<br>It can support multiple data centres for disaster recovery, redundancy and failover.                      |
| Scalability                                               | Easily scalable to various machines without any downtime or interruption to any applications.                                                 |
| Tunable consistency                                       | It offers tunable consistency for read and write operations.                                                                                  |
| MapReduce support                                         | It has Hadoop integration with support for MapReduce, Apache Pig and Apache Hive.                                                             |
| Query Language                                            | It introduces Cassandra Query Language.                                                                                                       |

### Benefits of Apache Cassandra

| Benefits                      | Description                                                                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scalability                   | Provides excellent stability to add customers and data as per the requirements.                                                                                                     |
| Architecture                  | Its architecture consists of nodes; data recovery is very easy in case of failures. It provides anytime access to data.                                                             |
| Fast linear scale performance | Enables fast response times due to linear scalability, i.e., it gets doubled with two nodes, quadruples with four, and so on.                                                       |
| Flexible data storage         | Easily accommodates the data in various formats such as structured, unstructured and semi-structured.<br>Changes can be made dynamically to the data structure as per requirements. |
| Easy data distribution        | Flexible distribution of data is provided across multiple servers, data centres and clouds.<br>Read and write operations to any node are automatically synchronised.                |
| Operational simplicity        | All the nodes are the same; hence there is no architectural complexity.                                                                                                             |
| Transaction support           | Through the use of the 'Commit log' feature, it is easy to capture all writes and redundancies, which ensures data durability in case of failures.                                  |

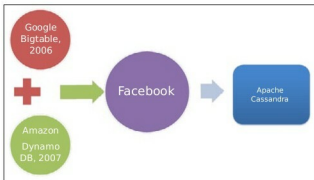


Figure 2: Apache Cassandra

data model. Rows are organised as a table, in which the first component of a table's primary key is the partition key. Within a partition, rows are clustered by the remaining columns of the key. Other columns may be indexed separately from the primary key. It does not support joins or sub-queries; rather, it emphasises de-normalisation through features such as collections. It supports linear scalability, which means that if two nodes can manage 100,000 operations per second, four nodes will be able to manage 200,000 operations per second and eight nodes will handle 400,000 operations per second.



Figure 3: Features of Cassandra

### Where can Apache Cassandra be used?

1. Cassandra can be used where there is a requirement for data to be ingested at a very high speed with all the features, reliability and redundancy.
2. It is used in situations where low coupling, high throughput